11 P is inversely proportional to the square root of m.

$P = 10$ when $m = \frac{1}{4}$

Work out the value of m when $P = 2$

$P = k \div \sqrt{m}$.

1 mark
Inverse square-root law

$10 \times \sqrt{(1/4)} = k$, so $k = 5$.

1 mark
Constant k=5

$2 = 5 \div \sqrt{m}$, so $\sqrt{m} = 5 \div 2$.

$m = (5 \div 2)^2 = 25 \div 4 = 6.25$.

1 mark
m=6.25

(Total for Question 11 is 3 marks)